M8-2SIO

Dual Serial Port for the Mark-8 Minicomputer

User's Guide

Rev A PC Boards

By Artem Kalinchuk

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Revision History

Manual	PCB	Date	Author	Notes

TABLE OF CONTENTS

Overview 4
Design Philosophy 4
Serial Port Configuration 5
Baud Rate Selection 5
Serial Port Address Selection 6
Programming the Serial Ports 7
Data Registers 7
Control Registers 7
Status Registers 9
Assembly and Test Instructions 11
General Assembly Tips 11
Assembly 11
Testing 16
Specifications 18
Appendix A: Software 19
PC Board Components 20
PC Board Drawings 22
Schematic Drawings 24

OVERVIEW

The M8-2SIO is a dual serial port board designed for the Mark-8 minicomputer. This board requires the M8-INMUX+ board in place of the original Input Multiplexer Board since it uses the bus for both input and output.

The M8-2SIO provides the following functions:

1. Two serial ports that can be independently enabled and disabled.
2. Configurable I/O ports for each serial port.
3. Configurable baud rate for each serial port.

Design Philosophy

The M8-2SIO uses only through-hole parts and ICs from the same era as the Mark-8 (circa 1974). The exception is a pair of MC34063 switching regulators (early 1980s), which generate ±12V from the Mark-8's +5V and -9V rails; so the board does not require external ±12V supply for RS-232. While we could use +5V and -9V for serial communication, some older terminals will not support those levels.

Jon Titus designed the Mark-8 around the Intel 8008 chip which supports 8 input ports and 24 output ports. That may seem like a sufficient number of ports but the problem is that the Mark-8's Input Multiplexer board limits the number of input ports to 2 since the board multiplexes the input ports and the CPU data. This is why the M8-INMUX+ board is required in place of the original Input Multiplexer board since it multiplexes the CPU data and all 8 of the supported input ports - via the bus. The output ports are already addressed and accessed via the bus.

One additional limitation of the original Mark-8 design is that the backplane is not uniform. There are 2 islands on the backplane which share all pins but pins 9-16. Those pins are shared amongst each board on each island but not between the islands. Because of this limitation, it's important that the M8-2SIO is placed in the "CPU/Mux" island so that pins 9-16 are shared with the CPU. This can be done by placing the M8-2SIO right after the CPU board.

While it may have been possible to leave the original Input Multiplexer board as-is and connect the M8-2SIO directly to the I/O ports of the Mark-8 (this design was used on the "clone" version of the Serial Board) the 2-input port limitation would still remain and the M8-2SIO would take up all of the available input ports and would only support 1 Serial Port (so it, in essence, it would be called the M8-1SIO).

In addition, the extra direct I/O connections would have created an unnecessary mess. Because of this, the decision was made to replace the Input Multiplexer board with the M8-INMUX+ - which enables the support of all 8 input ports and is a much cleaner design.

SERIAL PORT CONFIGURATION

The pins of the two 26-pin ribbon cable connectors (J1 and J2) are assigned such that a straight-through ribbon cable to a DB25S connector will provide normal RS-232 connections, as follows:

J1 or J2 Pin	DB25S Pin	Function	Direction
3	2	RS-232 RxD	In
5	3	RS-232 TxD	Out
7	4	RS-232 CTS	In
9	5	RS-232 RTS	Out
13	7	Ground	-

Baud Rate Selection

DIP Switch SW2 selects Port 0's baud rate, and DIP Switch SW3 sets Port 1's baud rate. Close the one switch that selects the desired baud rate for each port, and leave all other switches open.

SW2 or SW3 Position	Baud Rate
1	76800
2	38400
3	19200
4	9600
5	4800
6	2400
7	1200
8	600
9	300
10	110

These baud rates assume that the 6850 ACIAs are configured for ÷16 clocks. Additional baud rates can be achieved by configuring the ACIAs for ÷64 clocks

(discussed later). For example, you can get 150 baud by closing switch 8 and programming the ACIA for a ÷64 clock.

Note that baud rates over 19200 are not reliable at the speed of the Mark-8 if cycles are used to check the status register when each byte is received.

Serial Port Address Selection

Each Serial Port ACIA occupies two sequential I/O port addresses in the Mark-8. The first address is the port's control and status port; the second is the port's data port.

DIP switches SW1 and SW4 positions 2 through 3 set the serial port address, where closed represents a binary 0, and open represents a binary 1. Switch 2 is the highest-order bit, and switch 3 is the lowest. Switch 1 is used to enable and disable each individual Serial Port - closing this switch enables the port.

To configure Serial Port 0 to use address 000 (control/status) and address 001 (data), set SW2 to the following:

SW1 Position	Function	Setting
1	Enable	Closed
2	A2	Closed
3	A1	Closed

Notice that A0 is not configurable - this is because A0 is used by the software to specify whether to use the status/control register or the data register of the ACAIs.

The Mark-8 supports a maximum of 8 input ports (limited by the Intel 8008). Since each Serial Port requires 2 input ports and 2 output ports, the Mark-8 can, theoretically, support up to 4 individual Serial Ports (or 2 M8-2SIO boards).

PROGRAMMING THE SERIAL PORTS

The M8-2SIO's two serial ports are based on the Motorola MC6850 ACIA. See the Motorola MC6850 data sheet (DS9493R4, ©1995 Motorola) for details.

The base I/O address of the ACIAs is set by a DIP switch, as explained in the previous section. The M8-2SIO occupies two sequential I/O addresses for each Serial Port, starting at this base address. The first address is the control/status port; the second is data port. Each Serial Port address is configured individually.

A1	A0	Output Function	Input Function
0	0	Control Register	Status Register
0	1	Transmit Data Register	Receive Data Register

Data Registers

Data can be written to one of the Transmit Data Registers whenever the TDRE bit in the corresponding Status Register is high. Writing to the Transmit Data Register will clear that TDRE bit, which will remain clear until the ACIA transfers the data to its Transmit Shift Register for serialization.

The RDRF bit in one of the Status Registers will become high when the corresponding ACIA transfers a complete data word from its Receive Shift Register to its Receive Data Register, and will remain high until software reads from that Receive Data Register.

Control Registers

Each Port has an 8-bit control register that allows port configuration under software control. Each bit is defined as follows. (All bits are active-high.)

7	6	5	4	3	2	1	0
Input Interrupt	Output Interrupt		Transmission Bits			Clock Divide & Reset	

Bits 1 and 0 of each Control Register control the clock divider and master reset as follows:

Bit 1	Bit 0	Function
0	0	÷ by 1 clock

0	1	÷ by 16 clock
1	0	÷ by 64 clock
1	1	Master Reset

Software should first issue a Master Reset command to each Serial Port, by setting bits 1 and 0 of each control port to 1. For normal operation, the divide by 16 clock rate is then selected by writing 001 octal to the Serial Port's Control Register. Other (lower) baud rates are possible by selecting the divide by 64 clock rate - in which case the baud rate will be 1/4 of the baud rate printed on the PC board.

Bits 4, 3, and 2 of each Control Register set the word length, parity, and number of stop bits:

Data Bit			**Function**		
4	**3**	**2**	**# of Data Bits**	**# of Stop Bits**	**Parity**
0	0	0	7	2	Even
0	0	1	7	2	Odd
0	1	0	7	1	Even
0	1	1	7	1	Odd
1	0	0	8	2	None
1	0	1	8	1	None
1	1	0	8	1	Even
1	1	1	8	1	Odd

Bits 7, 6, and 5 of each Control Register control interrupts and handshaking:

Data Bit			**Function**
7	**6**	**5**	
X	0	0	RTS active, transmit interrupt disabled
X	0	1	RTS active, transmit interrupt enabled
X	1	0	RTS inactive, transmit interrupt disabled
X	1	1	RTS inactive, transmit interrupt disabled,

			transmits a BREAK on the transmit data line
0	X	X	Receive interrupt disabled
1	X	X	Receive interrupt enabled

The following 8008 code example illustrates initializing Port 0's ACIA (at its standard I/O address) for 8 data bits and 2 stop bits:

```
LAI    003        ;Master Reset
OUT0              ;...to the control port
LAI    021        ;8 data bits, 2 stop bits,
                   ;no parity, ÷ by 16 clock
OUT0              ;...to the control port
```

Status Registers

Each status register provides 8 status bits for the respective Serial Port, as follows:

Bit	Name	Function
7	IRQ	Interrupt Request
6	PE	Parity Error
5	OVRN	÷ by 64 clock
4	FE	Master Reset
3	/CTS	Clear to Send (active low)
2	/DCD	Data Carrier Detect (active low)
1	TDRE	Transmit Data Register Empty
0	RDRF	Receive Data Register Full

The IRQ bit will be high whenever the ACIA is requesting an interrupt. If transmit interrupts are enabled, then it will be high if the transmit data register is empty (and the TRDE bit is high). If receive interrupts are enabled, then the IRQ bit will be high if the receive data register is full (and the RDRF bit is high).

The three error conditions (PE, OVRN and FE) apply to the currently received data in the receive data register, and are valid while the RDRF bit is high.

The /CTS bit will be low whenever the RS-232 CTS signal is true, indicating that the ACIA may transmit data. When this bit is high, the RS-232 signal is false, and data transmission is inhibited by hardware within the ACIA.

The /DCD bit will be low whenever the RS-232 DCD signal is true, indicating that the ACIA may receive data. When this bit is high, the DCD signal is false, and data reception is inhibited by hardware within the ACIA.

A 0-to-1 transition of the /DCD bit generates an interrupt if the receive interrupt is enabled. This interrupt is cleared by first reading the Status Register, and then reading the Data Register, or by issuing a Master Reset command.

The TDRE and RDRF bits are discussed in the Data Registers section above.

Note on interrupts: Hardware interrupts are not yet supported.

ASSEMBLY AND TEST INSTRUCTIONS

General Assembly Tips

Install components in order of increasing height, beginning with the lowest-profile parts. This allows the work surface to hold the parts against the board when the assembly is inverted for soldering.

Refer to the PC Board Drawings (page 22) for the location of each component.

Install all components of a given type before proceeding to the next type (for example, all resistors, then all IC sockets). Secure the parts with low-adhesion tape (painter's tape) before inverting the board for soldering.

When installing DIP switches and sockets, seat each device fully against the board. Solder two diagonally opposite pins first. After all such devices are tacked in place, solder the remaining pins.

Warning: Some traces are very close to the solder pads. Use a small tip when soldering and be very careful to prevent solder shorts.

Assembly

Step 1 Install diodes and resistors

Component	Value	Location	Quantity
Diode, Schottky	1N5819	D1-D2	2
Resistor, 1/4W	330 Ω	R1-R2	2
Resistor, 1/4W	10K Ω	R3-R8	6
Resistor, 1/2W	0.33 Ω	R9,R14	2
Resistor, 1/4W	10.7K Ω	R10,R12	2
Resistor, 1/4W	1.24K Ω	R11,R13	2
Resistor, 1/4W	2.2K Ω	R15-R21	7

Refer to the PC Board Drawings (page 22) for the diode orientation.

Step 2 Install DIP sockets

Component	Location	Quantity

8-pin DIP socket, 0.3" wide	U14-U15	2
14-pin DIP socket, 0.3" wide	U1-U7,U11-U13	10
16-pin DIP socket, 0.3" wide	U10,U18-U19	3
20-pin DIP socket, 0.3" wide	U16-U17	2
24-pin DIP socket, 0.6" wide	U8-U9	2

See General Assembly Tips above. Tack each socket with two diagonally opposite pins, and complete soldering after the DIP switch has been similarly tacked in place.

Step 3 Install DIP switches

Component	Location	Quantity
DIP Switch, 3-Position SPST	SW1,SW4	2
DIP Switch, 10-Position SPST	SW2-SW3	2

Step 4 Install crystal

Component	Value	Location	Quantity
Crystal	4.9152 MHz	X1	1

The crystal can sit straight up but it’s recommended to lay it flat (by bending the leads of the crystal at a 90 degree angle) and then soldering a piece of wire to the body of the crystal (with the crystal value facing upwards) and the GND pad beside it.

Step 5 Install inductors

Component	Value	Location	Quantity
Inductor, Radial	220 uH, 1A	L1-L2	2

Step 6 Install ceramic capacitors

Component	Value	Location	Quantity
Capacitor, Ceramic	120 pF	C1	1
Capacitor, Ceramic	0.1 uF	C2-C11,C13-C18,C20-C21,C26-C29	22
Capacitor, Ceramic	470 pF	C12,C19	2

Step 7 Install electrolytic capacitors

Component	Value	Location	Quantity
Capacitor, electrolytic, 25 V, axial	220 uF	C22,C24	2
Capacitor, electrolytic, 25 V, axial	100 uF	C23	1
Capacitor, electrolytic, 25 V, axial	200 uF	C25	1

Observe polarity. The positive lead is marked "+" on the PC Board Drawings (page 22).

Step 8 Install connectors

Seat each connector flush with the board. Solder one pin at each end, verify alignment, then solder the remaining pins.

J3 is not a standard single-piece 41-position connector. Assemble it from standard Molex 0.156 in housings; for example, three 10-position connectors and one 11-position connector.

Component	Location	Quantity
Right-angle IDC header, 2x13	J1-J2	2
Right-angle female connector, Molex 0.156 in, 41-position	J3	1

Step 9 Inspect the assembly

1. Inspect all solder joints for omissions and bridges.
2. Use an ohmmeter to check for shorts between the following points. (Note that you will see measurable resistances because of the installed components, but not low-ohm short circuits.) **If you find any short circuits, do not continue until these are resolved!**

    a. Pin 3 and pin 6 of J3 (testing for +5V to ground shorts)
    b. Pin 3 and pin 7 of J3 (testing for -9V to ground shorts)
    c. Positive side of C22 and pin 3 of J3 (testing for +12V to ground shorts)
    d. Negative side of C24 and pin 3 of J3 (testing for -12V to ground shorts)
3. Insert U14 and U15 into their sockets. These are the MC34063A regulators.
4. Install the board into the Mark-8 backplane, power it on, and measure the following voltages, using pin 3 of J3 as the ground reference. It might help to run wires from the backplane to the board directly (pins 3, 6 and 7) to supply it with power while having it outside the Mark-8 for easier access. **If any of these voltages is incorrect, do not continue until they are corrected!**
    a. Pin 6 of J3 should measure between +4.9V and +5.2V
    b. Pin 7 of J3 should measure between -8.8V and +9.2V
    c. Positive side of C22 should measure between +11.8V and +12.2V
    d. Negative side of C24 should measure between -11.8V and -12.2V

Step 10 Clean the board

If organic-core or water-soluble-flux solder was used, cleaning is required. To remove water-soluble flux:

1. Wash the board after soldering is complete and before ICs are inserted. Hold the board under running hot water and scrub with a brush for several minutes.
2. Remove water with compressed air, including from beneath components and from inside IC sockets. A workshop compressor is more effective than aerosol dusters.
3. Dry the board near a moderate heat source for approximately 30 minutes.

If rosin-core solder was used, cleaning is optional. Apply a flux remover suitable for rosin; a brush may be used to loosen residue. Inspect the solder joints again after cleaning.

Step 11 Install ICs

Insert each IC with pin 1 toward the top of the board, as shown on the PC Board Drawings. Do not fold pins under the body. If necessary, press each row of pins against the work surface to set the lead pitch before insertion.

Refer to the PC Board Drawings for IC locations and orientation.

Component	Value	Location	Quantity
IC, DIP, Quad RS232 Line Driver	1488	U1,U6	2
IC, DIP, Quad RS232 Line Receiver	1489	U2,U7	2

IC, DIP, 4-Bit Ripple Counter	74LS393	U3,U5	2
IC, DIP, Quad 2-Input NAND Gate	74LS00	U4,U13	2
IC, DIP, ACIA	MC6850	U8-U9	2
IC, DIP, 4-Bit Loadable Counter	74LS161	U10	1
IC, DIP, Quad 2-input OR Gate	74LS32	U11	1
IC, DIP, Hex Inverter	74LS04	U12	1
IC, DIP, Switching Regulator	MC34063A	U14-U15	2
IC, DIP, Octal D Tri-State Latch, Level Triggered	74LS373	U16	1
IC, DIP, Octal D Tri-State Latch, Edge Triggered	74LS374	U17	1
IC, DIP, 4-bit Magnitude Comparator	74LS85	U18-U19	2

Step 12 Configuration

Set the DIP switches as follows. 0 means closed, 1 means open.

Switch	Setting	Switch Function
SW1<2:1>	00	Serial Port 0 address
SW1<3>	Closed	Serial Port 0 enabled
SW2<10:1>	As needed	Serial Port 0 baud rate (I suggest 9600 baud.)
SW3<10:1>	As needed	Serial Port 1 baud rate
SW4<2:1>	01	Serial Port 1 address
SW4<3>	Closed	Serial Port 1 enabled

Note that SW2 and SW3 can only have one switch closed at a time. These switches are used to select the baud rate of each Serial Port.

Step 13 Final inspection

Inspect the completed board for the following defects:

- IC installed with reversed orientation
- IC pin folded under the package
- Incorrect device in any socket
- ICs not fully seated

- Incorrect switch setting

Testing

The following procedure verifies basic operation of the M8-2SIO in a working Mark-8 system. It is not a complete troubleshooting guide. Diagnosis of a non-functional system is beyond the scope of this manual.

Step 1 Installation

As noted in the Overview, the M8-2SIO PCB requires the modified Input Multiplexer PCB (M8-INMUX+) in place of the original Input Multiplexer board. This is because the original Input Multiplexer did not support data input from the bus and had a maximum of 2 input ports.

The Mark-8 backplane has two separate islands - the CPU/Mux island and the Memory/Output island. These two islands are separated at pins 9-16 of the backplane. Insert the M8-2SIO PCB into the CPU/Mux island so that pins 9-16 of the backplane are shared with the CPU, M8-INMUX+ and the M8-2SIO. This can simply be done by inserting the M8-2SIO right after the CPU board.

Step 2 Internal cabling

Install cables from the M8-2SIO to the rear panel of the computer. Use a standard 26-pin ribbon cable to DB25S connector. (This type of cable was commonly used on PCs for the parallel port, and are still readily available in several lengths.)

Step 3 External cabling

You will need a cable from the rear of the Mark-8 to whatever terminal you will use for testing. If you are using a terminal emulator on a PC, then you probably have a standard male 9-pin connector for serial communication, requiring a cable like this. (The other pins don’t matter.) This is a standard cable that you can either buy or make.

M8-2SIO Signal	Mark-8 DB25 (Male)	Direction	PC DB9S (Female)
TxD	3	→	2
RxD	2	←	3
GND	7	-	5

Connect this cable to the M8-2SIO’s Port 0 and to your terminal or PC.

Step 4 Setting up your terminal or terminal program

Set up your terminal or terminal program for the same baud rate as the M8-2SIO's port 0 (i.e. 9600). Select full duplex, no handshaking, 8 bit data, no parity, 2 stop bits. If you are using a terminal emulator on a PC, be sure to select the correct COM port.

Step 5 Testing ASCII echo

Power on the Mark-8 and enter the program titled "ASCII Echo" (found under Appendix A) into the Mark-8 via the front panel switches. Be sure to start at address 0.

Set the Mark-8 into "RUN" and "INT" modes, select 005 on the Switch Register and execute an interrupt by toggling the interrupt switch. The Mark-8 should start listening for data on Serial Port 0. If the computer halts right away, ensure the program was entered correctly.

On the terminal or terminal emulator on a PC, type the character "A". If everything is working correctly, the character that was typed should be echoed back to the terminal and displayed to the user. If not, check the following:

- Port 0 baud rate setting matches the terminal baud rate
- Double-check all switch settings (SW1 should have all switches closed)
- Serial cable connections are correct
- Baud clock working correctly (use a scope to check pin 3 or 4 of U8)

If the ASCII echo works fine, try testing the same program on Serial Port 1. Be sure to disable Serial Port 0 (SW1<3> Open), enable Serial Port 1 (SW4<3:1> all closed) and swap the cabling to Serial Port 1.

SPECIFICATIONS

PC Board

Material:	FR-4 material, no solder mask, tin finish
Thickness	0.062” +/- 0.006”
Silkscreen:	None
Layers:	2

Serial Ports

Serial Ports:	2
UART Type	Motorola MC6850
Baud Rates:	110, 300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 76800 (DIP switch selectable)
Protocol:	RS232
Handshaking:	RS-232 CTS, RTS
Connectors:	26-pin header

APPENDIX A: SOFTWARE

ASCII Echo

This code will listen to input on Serial Port 0 and, once received, will echo back the data that is received.

```
RDA     EQU     001
TBE     EQU     002
        ORG     000

; INIT
        LAI     003             ;006 003        - master reset
        OUT0                    ;121
        LAI     021             ;006 021        - ÷16, 8N2
        OUT0                    ;121

; WAIT FOR RX
INPUT:
        IN0                     ;101            - status
        LBA                     ;310
        LAI     RDA             ;006 001
        NDB                     ;241
        JTZ     INPUT           ;150 006 000
        IN1                     ;103            - data
        LCA                     ;320

; WAIT FOR TX THEN ECHO
OUTPUT:
        IN0                     ;101            - status
        LBA                     ;310
        LAI     TBE             ;006 002
        NDB                     ;241
        JTZ     OUTPUT          ;150 020 000
        LAC                     ;302
        OUT1                    ;123            - data
        JMP     INPUT           ;104 006 000
        END
```

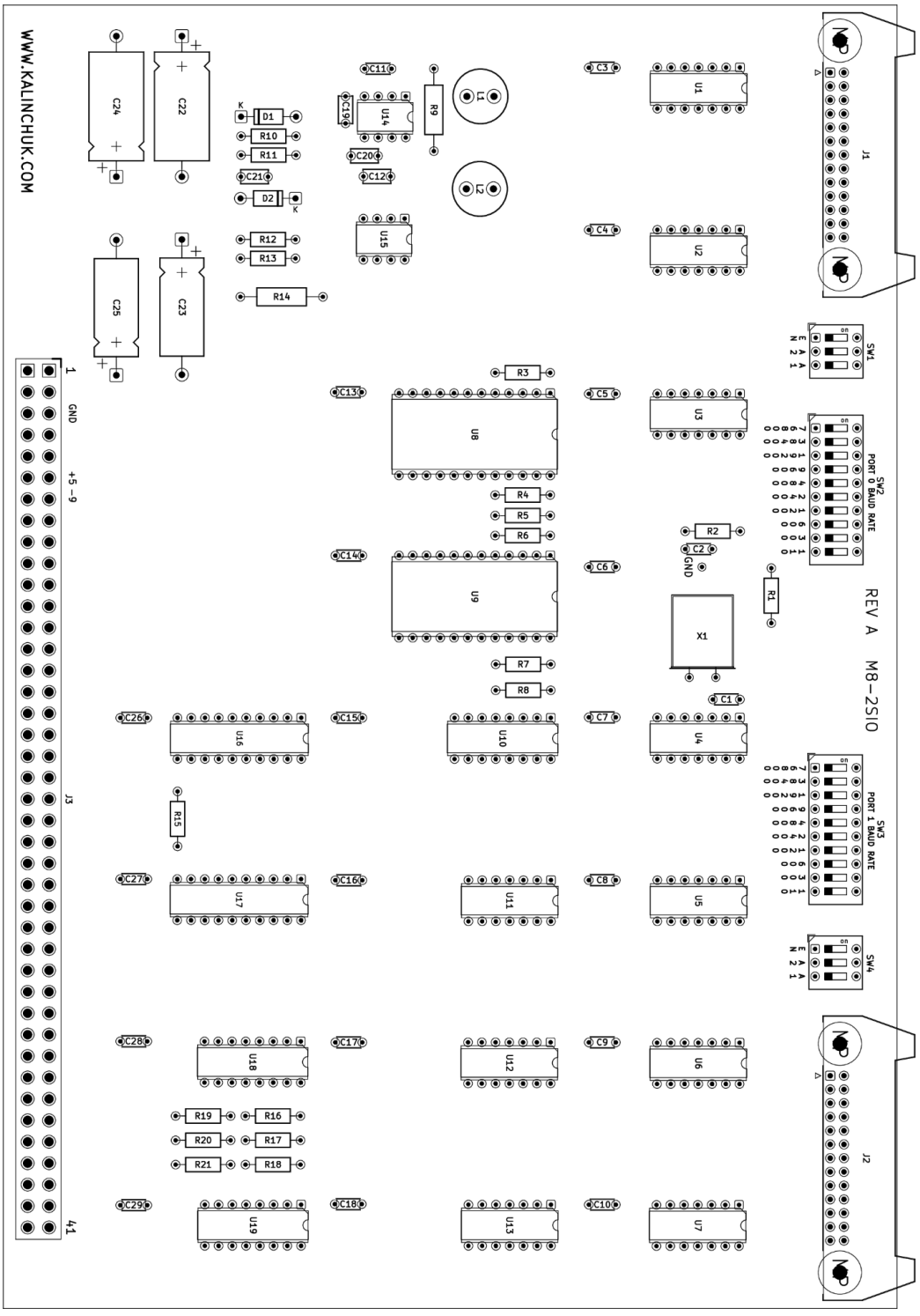
PC BOARD COMPONENTS

Component	Value	Location	Quantity
PC Board, M8-2SIO, Rev A			1
Diode, Schottky	1N5819	D1-D2	2
Resistor, 1/4W	330 Ω	R1-R2	2
Resistor, 1/4W	10K Ω	R3-R8	6
Resistor, 1/2W	0.33 Ω	R9,R14	2
Resistor, 1/4W	10.7K Ω	R10,R12	2
Resistor, 1/4W	1.24K Ω	R11,R13	2
Resistor, 1/4W	2.2K Ω	R15-R21	7
8-pin DIP socket, 0.3" wide		U14-U15	2
14-pin DIP socket, 0.3" wide		U1-U7,U11-U13	10
16-pin DIP socket, 0.3" wide		U10,U18-U19	3
20-pin DIP socket, 0.3" wide		U16-U17	2
24-pin DIP socket, 0.6" wide		U8-U9	2
DIP Switch, 3-Position SPST		SW1,SW4	2
DIP Switch, 10-Position SPST		SW2-SW3	2
Crystal	4.9152 MHz	X1	1
Inductor, Radial	220 uH, 1A	L1-L2	2
Capacitor, Ceramic	120 pF	C1	1
Capacitor, Ceramic	0.1 uF	C2-C11,C13-C18,C20-C21,C26-C29	22
Capacitor, Ceramic	470 pF	C12,C19	2
Capacitor, electrolytic, 25 V, axial	220 uF	C22,C24	2
Capacitor, electrolytic, 25 V, axial	100 uF	C23	1
Capacitor, electrolytic, 25 V, axial	200 uF	C25	1
Right-angle IDC header, 2x13		J1-J2	2

Right-angle female connector, Molex 0.156 in, 41-position		J3	1
IC, DIP, Quad RS232 Line Driver	1488	U1,U6	2
IC, DIP, Quad RS232 Line Receiver	1489	U2,U7	2
IC, DIP, 4-Bit Ripple Counter	74LS393	U3,U5	2
IC, DIP, Quad 2-Input NAND Gate	74LS00	U4,U13	2
IC, DIP, ACIA	MC6850	U8-U9	2
IC, DIP, 4-Bit Loadable Counter	74LS161	U10	1
IC, DIP, Quad 2-input OR Gate	74LS32	U11	1
IC, DIP, Hex Inverter	74LS04	U12	1
IC, DIP, Switching Regulator	MC34063A	U14-U15	2
IC, DIP, Octal D Tri-State Latch, Level Triggered	74LS373	U16	1
IC, DIP, Octal D Tri-State Latch, Edge Triggered	74LS374	U17	1
IC, DIP, 4-bit Magnitude Comparator	74LS85	U18-U19	2

PC BOARD DRAWINGS

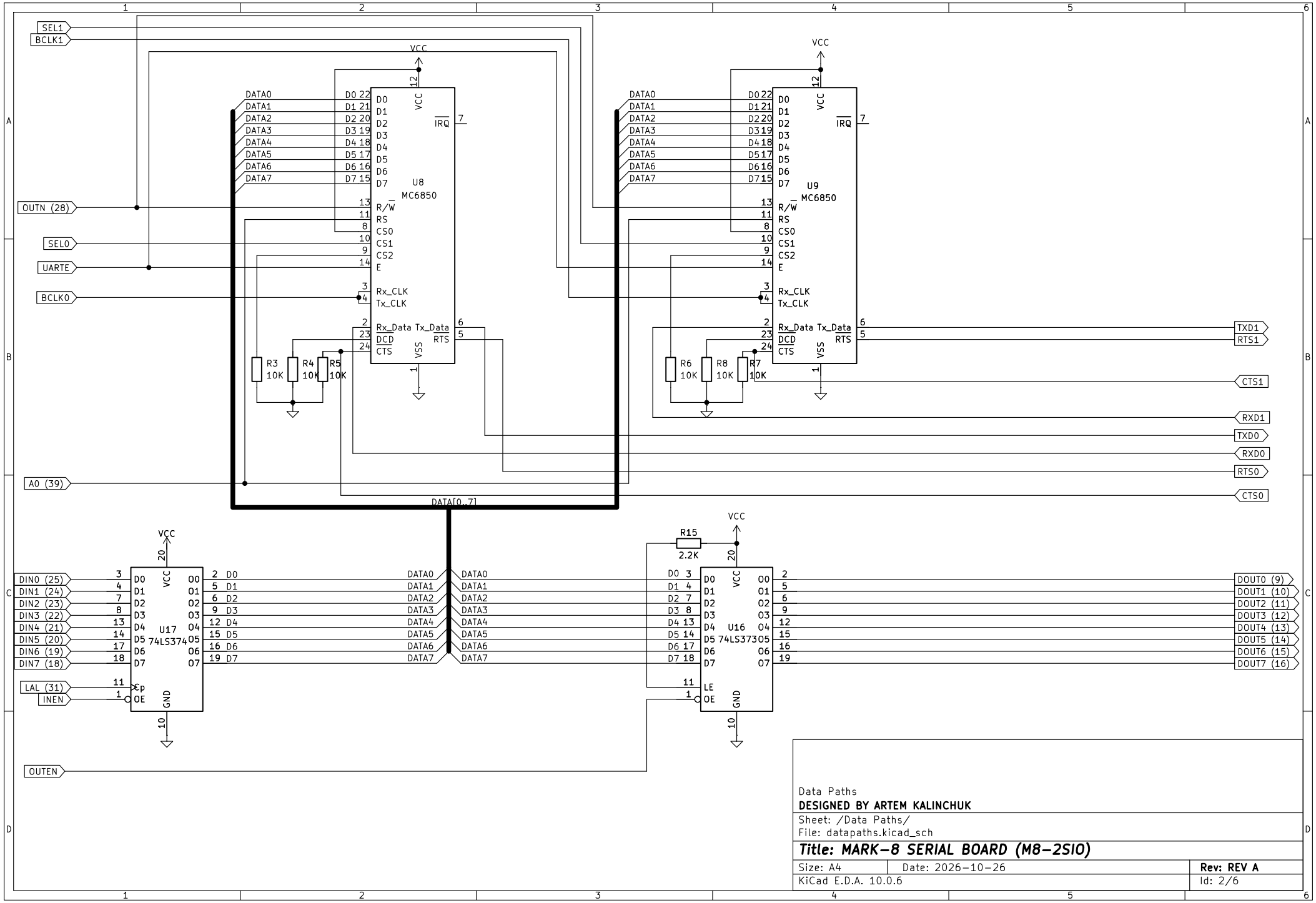
The following pages show component placement for Rev A.



Rev A Component Placement

SCHEMATIC DRAWINGS

The following pages contain the schematic for Rev A.



Data Paths

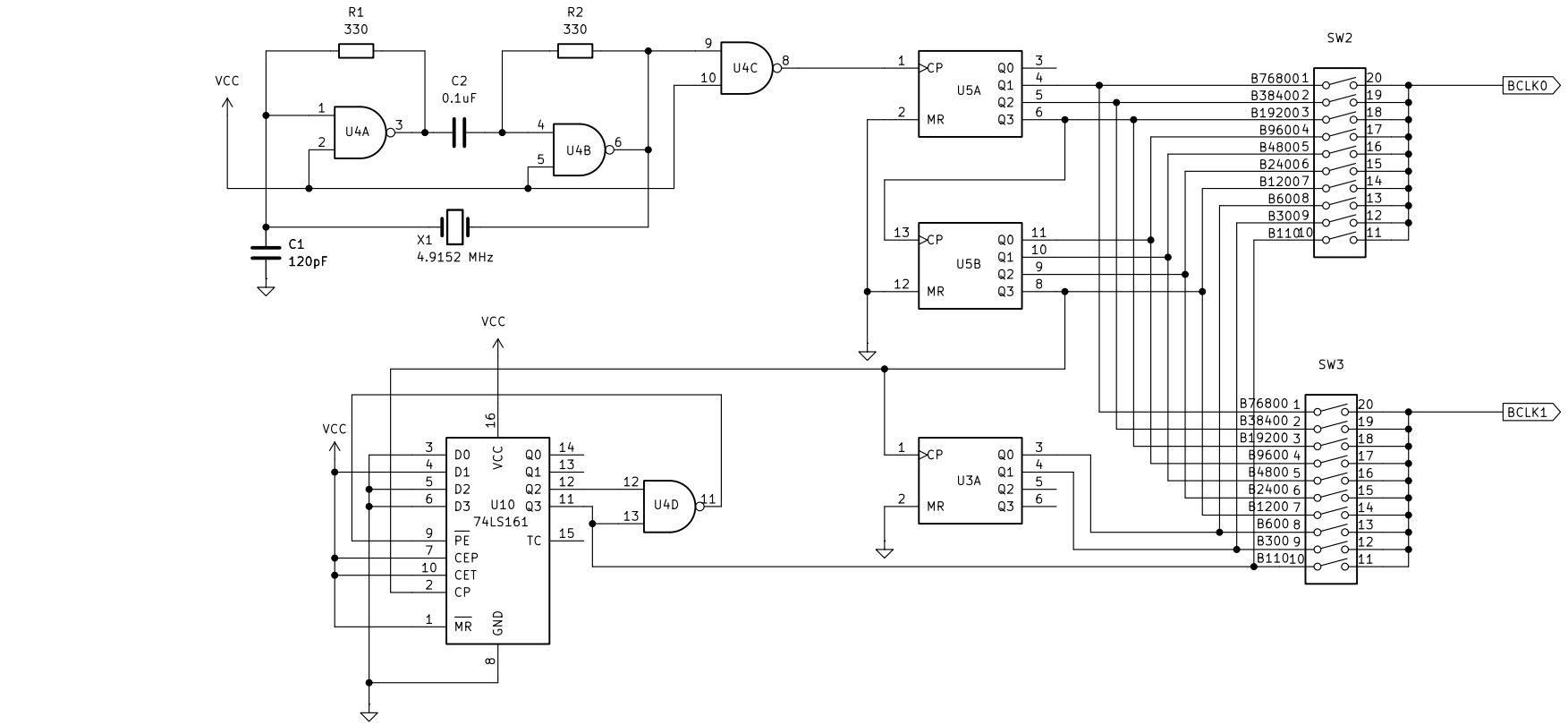
DESIGNED BY ARTEM KALINCHUK

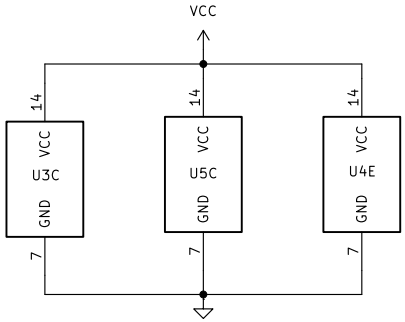
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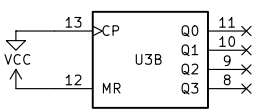
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Size: A4	Date: 2026-10-26	**Rev: REV A**
KiCad E.D.A. 10.0.6		Id: 2/6





U3B unused



Baud Clock

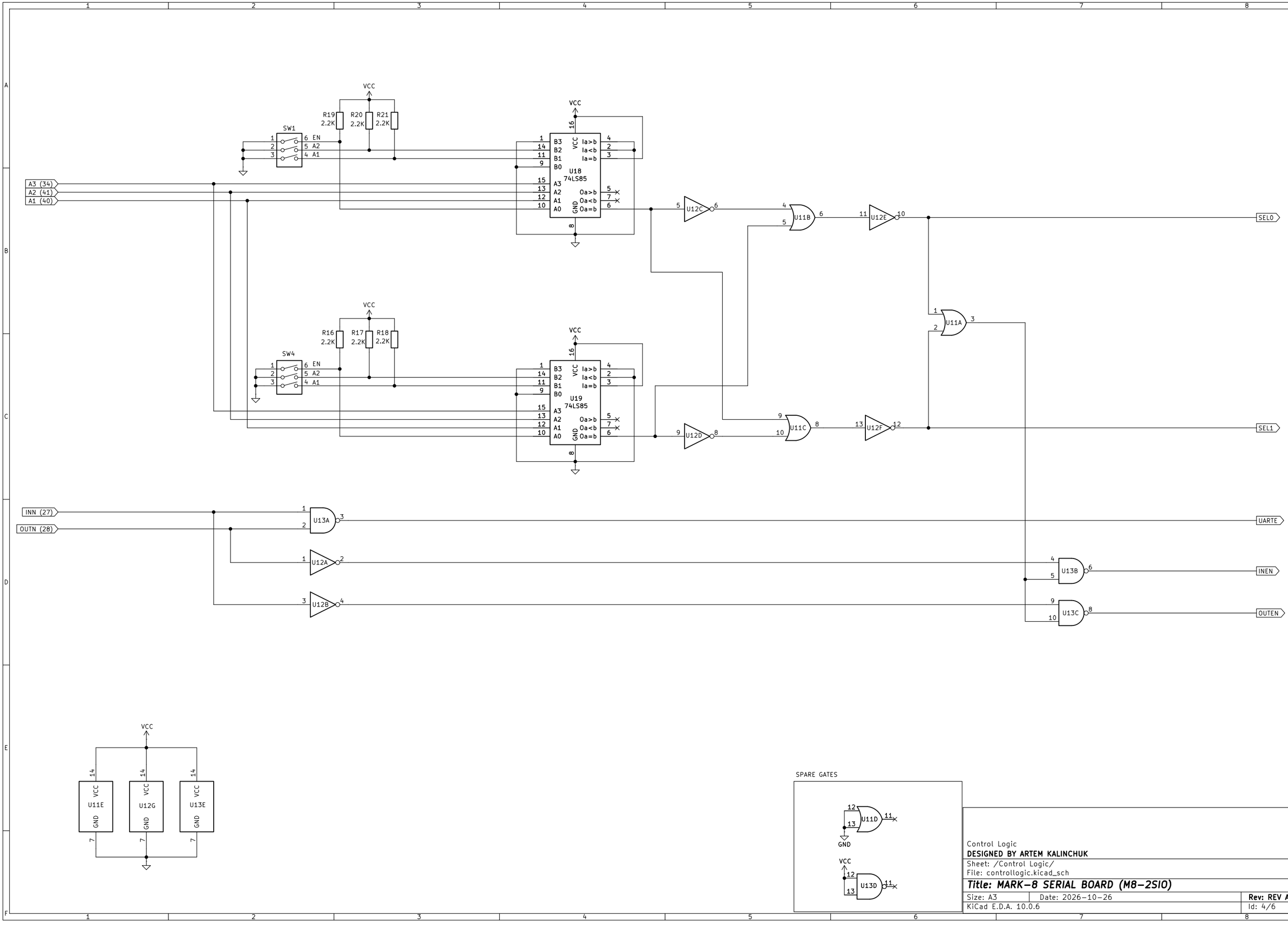
DESIGNED BY ARTEM KALINCHUK

Sheet: /Baud Clock/

File: baudclock.kicad_sch

Title: MARK-8 SERIAL BOARD (M8-2SIO)

Size: A4	Date: 2026-10-26	**Rev: REV A**
KiCad E.D.A. 10.0.6		Id: 3/6



Control Logic

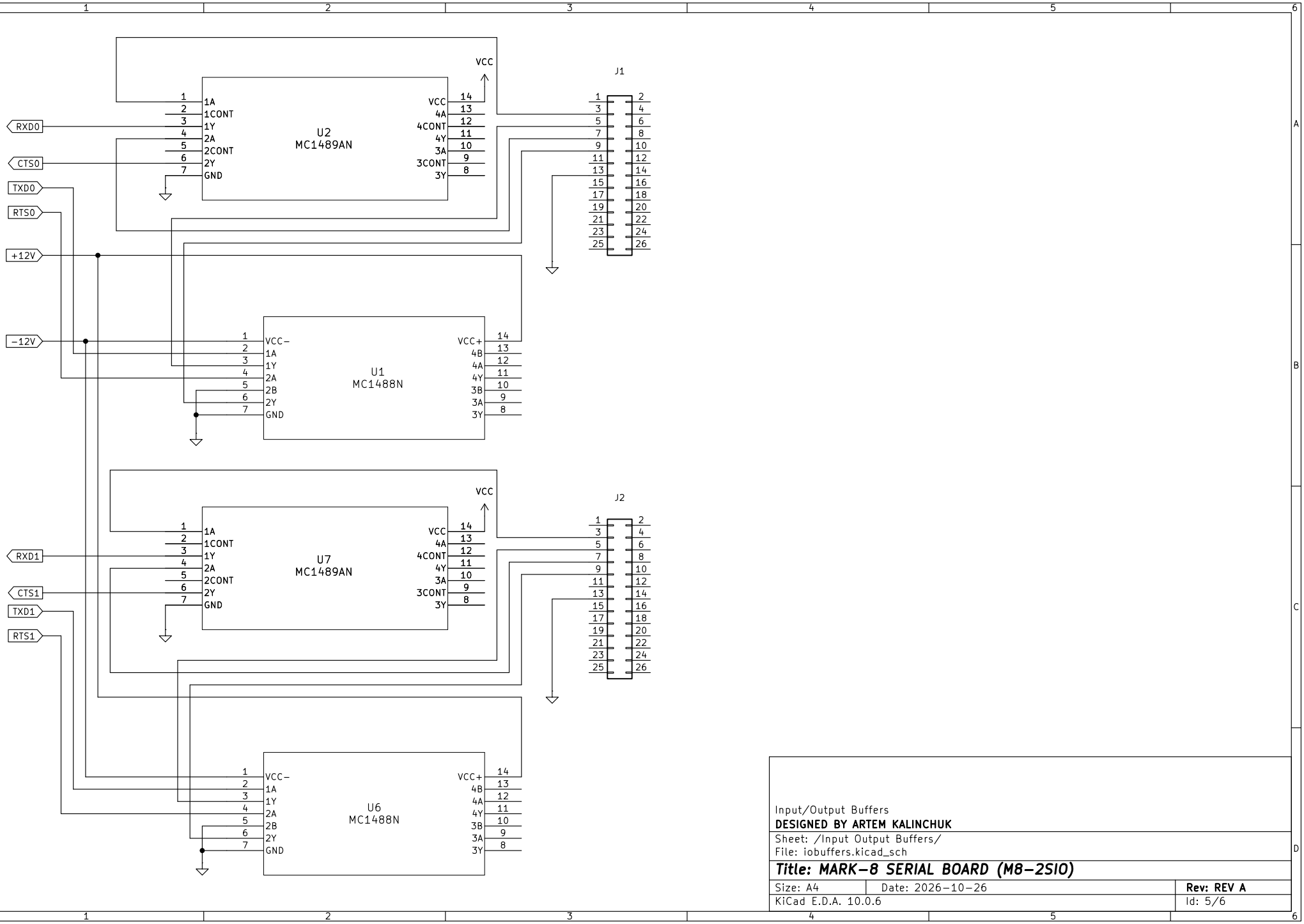
DESIGNED BY ARTEM KALINCHUK

Sheet: /Control Logic/

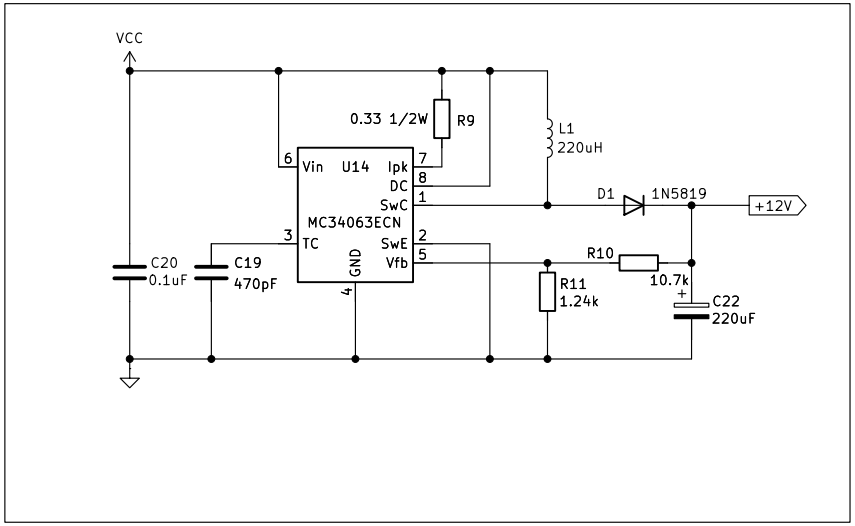
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Title: MARK-8 SERIAL BOARD (M8-2SIO)

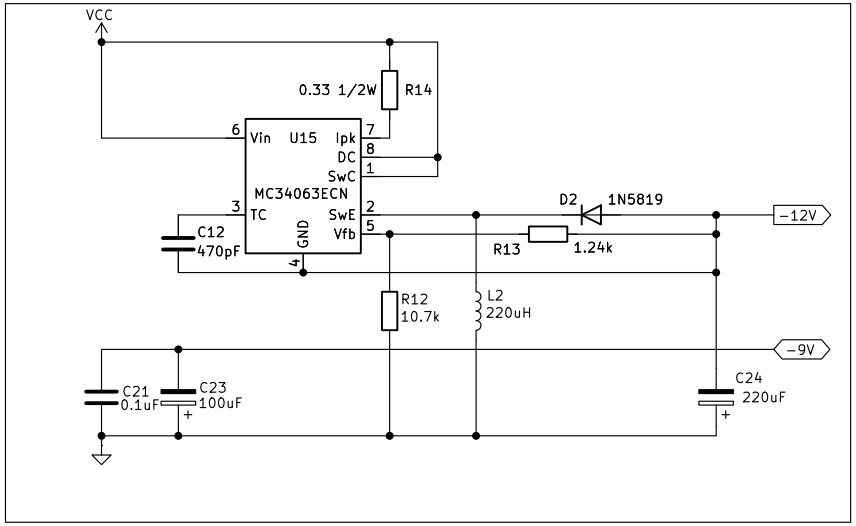
Size: A3	Date: 2026-10-26	**Rev: REV A**
KiCad E.D.A. 10.0.6		Id: 4/6


U2
MC1489AN
1A
1CONT
1Y
2A
2CONT
2Y
GND
VCC
4A
4CONT
4Y
3A
3CONT
3Y
RXD0
CTS0
TXD0
RTS0
+12V
−12V
J1
U1
MC1488N
VCC−
VCC+
4B
3B
U7
MC1489AN
RXD1
CTS1
TXD1
RTS1
J2
U6
MC1488N
Input/Output Buffers
DESIGNED BY ARTEM KALINCHUK
Sheet: /Input Output Buffers/
File: iobuffers.kicad_sch
Title: MARK−8 SERIAL BOARD (M8−2SIO)
Size: A4
Date: 2026−10−26
Rev: REV A
KiCad E.D.A. 10.0.6
Id: 5/6

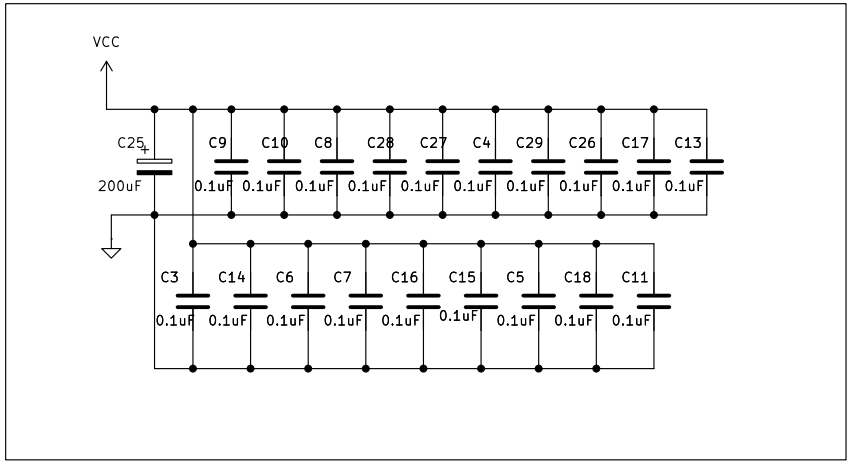
+5V TO +12V



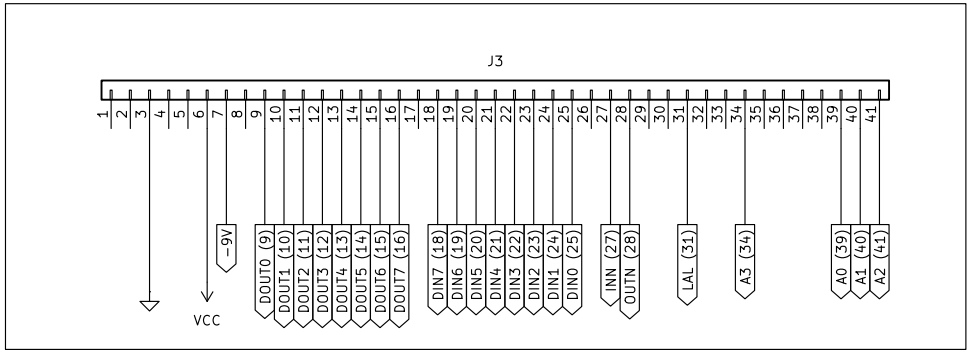
−9V TO −12V



DECOUPLING CAPACITORS



BUS CONNECTIONS



Connections/Power

DESIGNED BY ARTEM KALINCHUK

Sheet: /Connections / Power/

File: connections.kicad_sch

Title: MARK−8 SERIAL BOARD (M8−2S10)

Size: A4	Date: 2026−10−26	**Rev: REV A**
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